4.4.4 Stationary waves

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	stationary (standing) waves using microwaves, stretched strings and air columns	
(b)	graphical representations of a stationary wave	
(c)	similarities and the differences between stationary and progressive waves	
(d)	nodes and antinodes	
(e)	**(i)** stationary wave patterns for a stretched string and air columns in closed and open tubes	**PAG5**
	(ii) techniques and procedures used to determine the speed of sound in air by formation of stationary waves in a resonance tube	
(f)	the idea that the separation between adjacent nodes (or antinodes) is equal to $\lambda/2$, where λ is the wavelength of the progressive wave	
(g)	fundamental mode of vibration (1st harmonic); harmonics.	